AN ADEQUATE MODULE IS QUASI-COHERENT ON THE ZARISKI SITE, VANISHES ON AFFINES AND COMPUTES THE SAME COHOMOLOGY

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Allowed standard prerequisites (editorial).

- The flat base change property of adequate modules and their locality. Same chapter, Lemmas lemma-adequate-flat and lemma-adequate-local, invoked at proof lines 1914-1921. Give F(U) = F(V) tensor O(U) for flat U -> V, hence quasi-coherence of the Zariski restriction.
- Cohomology of a quasi-coherent module on an affine scheme vanishes in positive degrees, and the comparison between Zariski and site cohomology. The standard Cech computation for quasi-coherent modules, used in the closing paragraph of the proof. Gives the vanishing and the comparison statement.

Source reference. Original lemma 03FI: sources/stacks-chow/tags.txt, source lines 4493–4493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03DW: sources/stacks-chow/tags.txt, source lines 4435–4435. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06UW: sources/stacks-adequate/pinned-original-source.tex, source lines 442–442. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VL: sources/stacks-adequate/pinned-original-source.tex, source lines 1841–1841. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 06VI: sources/stacks-adequate/pinned-original-source.tex, source lines 1707–1707. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03F9: sources/stacks-chow/tags.txt, source lines 4484–4484. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be an adequate $\mathcal{O}$-module on $(Sch/S)_\tau$.*

1. *The restriction $\mathcal{F}|_{S_{Zar}}$ is a quasi-coherent $\mathcal{O}_S$-module on the scheme S.*
2. *The restriction $\mathcal{F}|_{S_{étale}}$ is the quasi-coherent module associated to $\mathcal{F}|_{S_{Zar}}$.*
3. *For any affine scheme U over S we have $H^q(U, \mathcal{F}) = 0$ for all $q > 0$.*
4. *There is a canonical isomorphism*

$$H^q(S, \mathcal{F}|_{S_{Zar}}) = H^q((Sch/S)_\tau, \mathcal{F}).$$

Proof. By Lemma 06UW and Lemma 06VI we see that for any flat morphism of affines $U \to V$ over S we have $\mathcal{F}(U) = \mathcal{F}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U)$. This works in particular if $U \subset V \subset S$ are affine opens of S, hence $\mathcal{F}|_{S_{Zar}}$ is quasi-coherent. Thus (1) holds.

Let $S' \to S$ be an étale morphism of schemes. Then for $U \subset S'$ affine open mapping into an affine open $V \subset S$ we see that $\mathcal{F}(U) = \mathcal{F}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U)$ because $U \to V$ is étale, hence flat. Therefore $\mathcal{F}|_{S'_{Zar}}$ is the pullback of $\mathcal{F}|_{S_{Zar}}$. This proves (2).

We are going to apply Cohomology on Sites, Lemma 03F9 to the site $(Sch/S)_\tau$ with $\mathcal{B}$ the set of affine schemes over S and Cov the set of standard affine τ-coverings. Assumption (3) of the lemma is satisfied by Descent, Lemma 03FI and Lemma 06VL for the case of a covering by a single affine. Hence we conclude that $H^p(U, \mathcal{F}) = 0$ for every affine scheme U over S. This proves (3). In exactly the same way as in the proof of Descent, Proposition 03DW this implies the equality of cohomologies (4). □